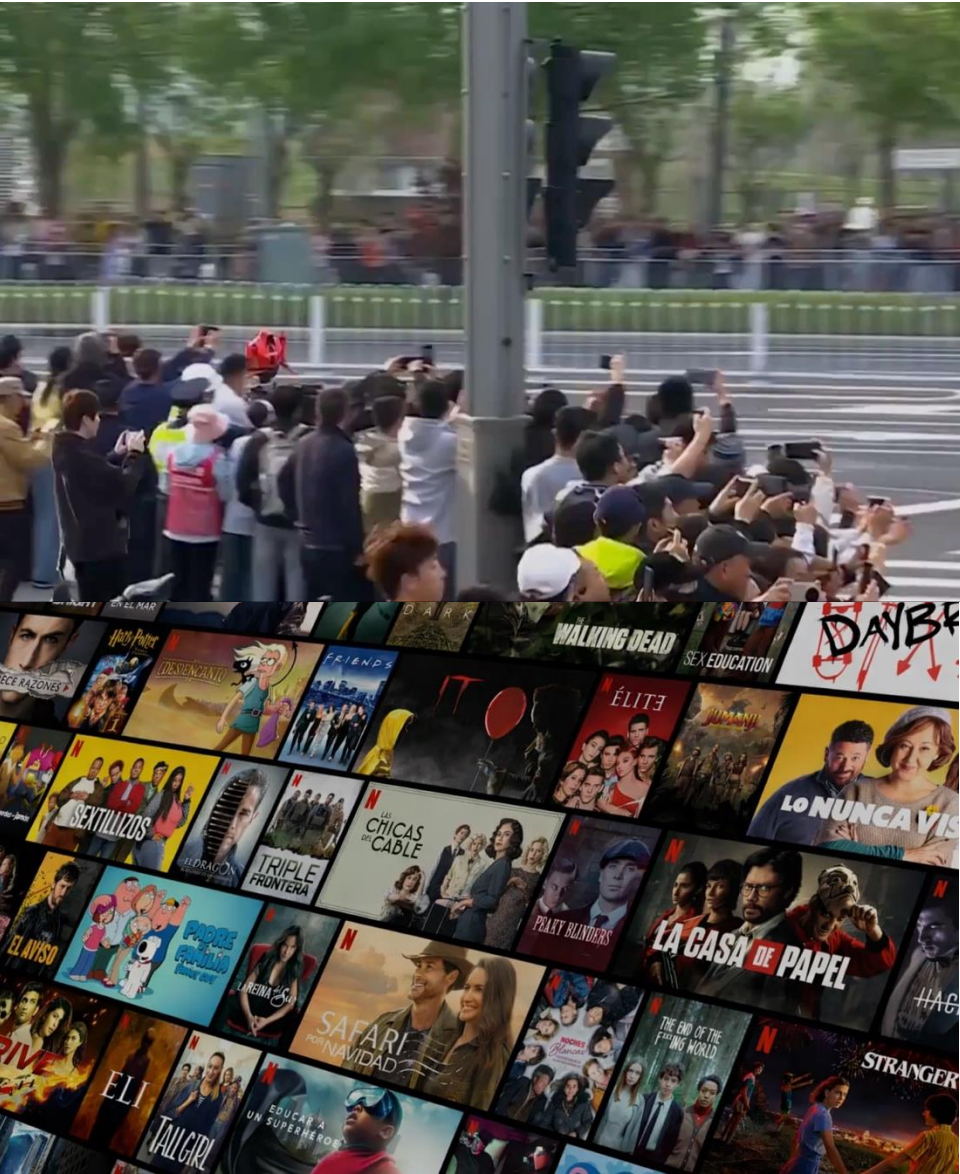


Intro to Optimization Under Uncertainty

**2026 Fall
AD7031**



AI Changes World



ChatGPT

Gemini



perplexity

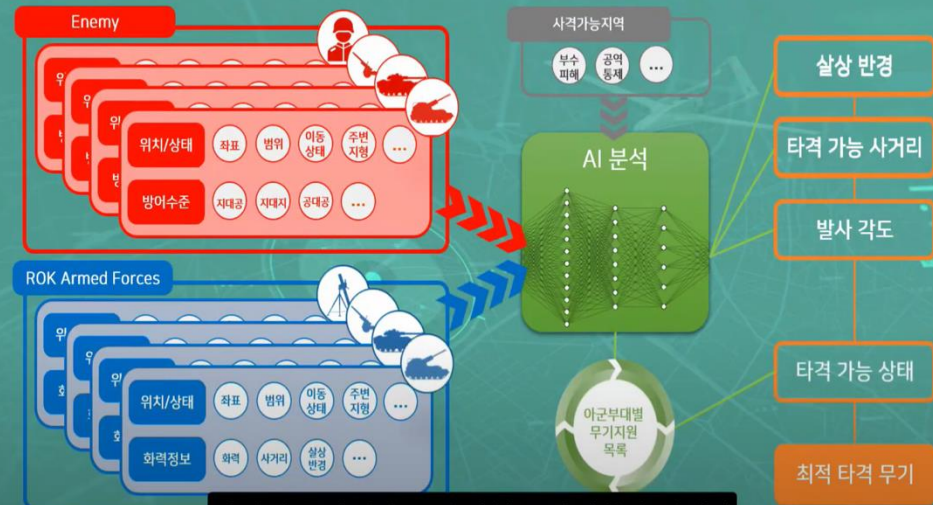


Claude

AI in Defense



METT-TC 기반 AI 분석



최적의 공격킬체인 구성 방식을 제안하게 됩니다



원료 또는 재료의 50% 넘게 사용하여 가공한 제품만 해당한다.

2. [검색된 정보를 바탕으로 생각해볼게요]

- 사용자는 설명절 기간에 허용되는 선물의 종류를 묻고 있습니다.
- 지식베이스에서 설명절 기간 동안 허용되는 선물의 가액 범위와 농수산물 선물의 종류에 대한 정보를 찾았습니다.
- 농수산물 및 농수산물가공품 선물은 30만원까지 허용되며, 농수산물에는 농산물, 수산물, 축산물, 임산물 이 포함됩니다.

3. [최종 답변] 요청하신 질문은 설명절 기간에 허용되는 선물의 종류는 무엇인가요? 설명절 기간 동안 공직자에

GeDAI (Generative Defense AI)

국방 생성형 인공지능

Why Robust AI?



Boston Dynamics Atlas



Tesla Autopilot

Why Robust AI?

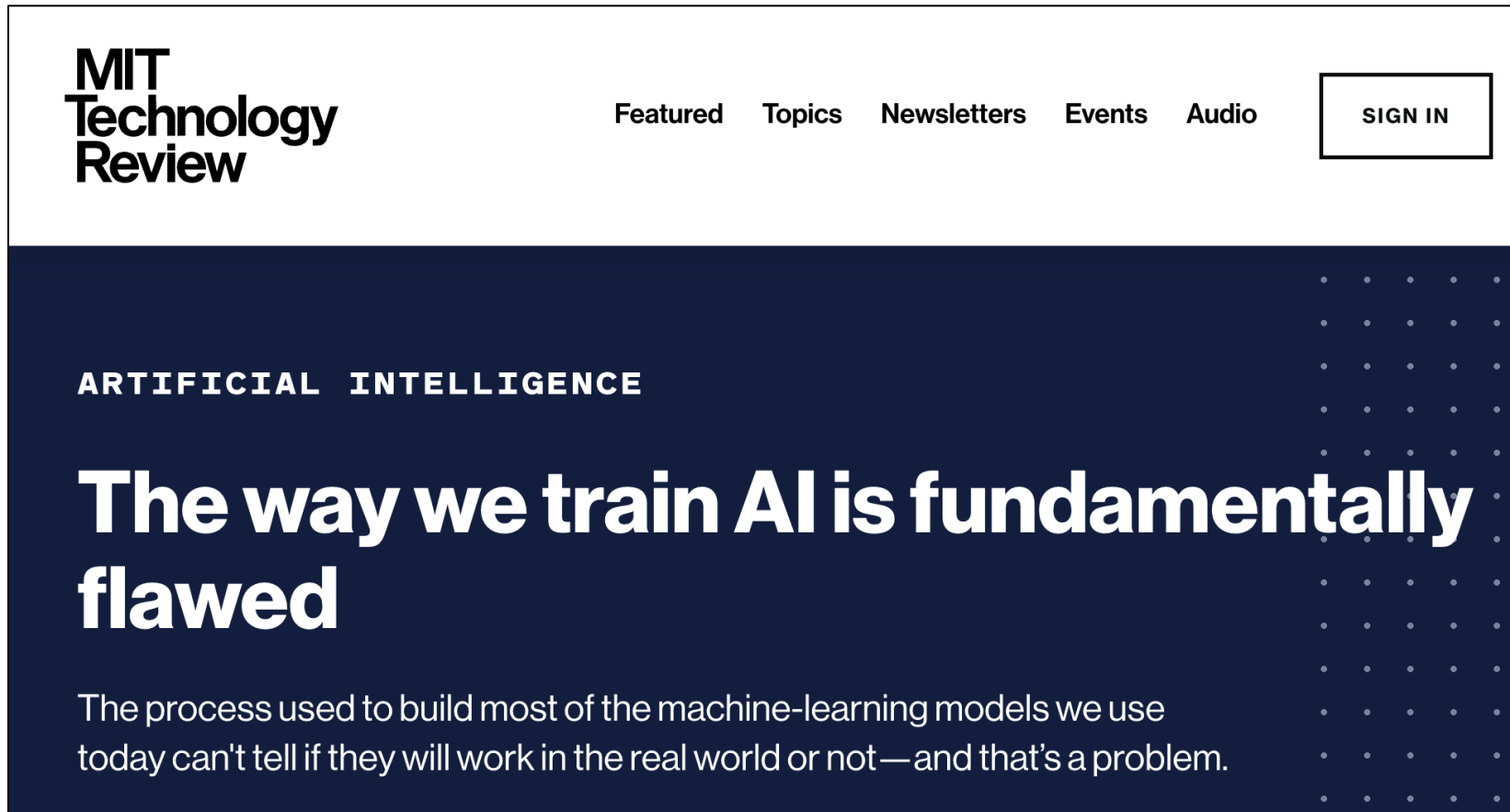


Boston Dynamics Atlas



Tesla Autopilot

Why Robust AI?

The image shows the header of an MIT Technology Review article. It features a dark blue background with a grid of small white dots on the right side. The MIT Technology Review logo is in the top left, and navigation links are in the top right. The article title is in large white text, and the introductory paragraph is below it.

**MIT
Technology
Review**

[Featured](#) [Topics](#) [Newsletters](#) [Events](#) [Audio](#) [SIGN IN](#)

ARTIFICIAL INTELLIGENCE

The way we train AI is fundamentally flawed

The process used to build most of the machine-learning models we use today can't tell if they will work in the real world or not—and that's a problem.

AI is not magic —
it is optimization under uncertainty



ChatGPT

Gemini



perplexity



Claude



Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $y \in \mathbb{R}^n$
- Classification : predict binary $y \in \{-1, +1\}$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $y \in \mathbb{R}^n$

$$\text{Squared Loss}(\cdot) = \|\cdot\|_2^2 = \|y - f_{\theta}(x)\|_2^2$$

e.g., when $n=2$

$$y_1 = \begin{bmatrix} 100 \\ 10 \end{bmatrix}, f_{\theta}(x_1) = \begin{bmatrix} 105 \\ 9 \end{bmatrix}$$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $\mathbf{y} \in \mathbb{R}^n$

$$\text{Squared Loss}(\cdot) = \|\cdot\|_2^2 = \|\mathbf{y} - f_{\theta}(\mathbf{x})\|_2^2$$

e.g., when $n=2$

$$\mathbf{y}_1 = \begin{bmatrix} 100 \\ 10 \end{bmatrix}, f_{\theta}(\mathbf{x}_1) = \begin{bmatrix} 105 \\ 9 \end{bmatrix}$$

$$\Rightarrow \left\| \begin{bmatrix} 100 \\ 10 \end{bmatrix} - \begin{bmatrix} 105 \\ 9 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} -5 \\ 1 \end{bmatrix} \right\|_2^2 = (-5)^2 + 1^2 = 26$$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $\mathbf{y} \in \mathbb{R}^n$

$$\text{Squared Loss}(\cdot) = \|\cdot\|_2^2 = \|\mathbf{y} - f_{\theta}(\mathbf{x})\|_2^2$$

e.g., when $n=1$

$$y_1 = 10, f_{\theta}(x_1) = 11$$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $y \in \mathbb{R}^n$

$$\text{Squared Loss}(\cdot) = \|\cdot\|_2^2 = \|y - f_{\theta}(x)\|_2^2$$

e.g., when $n=1$

$$\begin{aligned} y_1 &= 10, f_{\theta}(x_1) = 11 \\ \Rightarrow \|10 - 11\|_2^2 &= \|-1\|_2^2 = (-1)^2 = 1 \end{aligned}$$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Classification : predict binary $y \in \{-1, +1\}$

$$\mathbf{0-1 Loss}(\cdot) = \begin{cases} 0, & \text{if } \text{sign}(f_{\theta}(x)) = \text{sign}(y) \\ 1, & \text{if } \text{sign}(f_{\theta}(x)) \neq \text{sign}(y) \end{cases}$$

e.g., $y_1 = -1, f_{\theta}(x_1) = -1000.5$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Classification : predict binary $y \in \{-1, +1\}$

$$\text{0-1 Loss}(\cdot) = \begin{cases} 0, & \text{if } \text{sign}(f_{\theta}(x)) = \text{sign}(y) \\ 1, & \text{if } \text{sign}(f_{\theta}(x)) \neq \text{sign}(y) \end{cases}$$

e.g., $y_1 = -1, f_{\theta}(x_1) = -1000.5 \Rightarrow \text{Correct! 0-loss}$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Classification : predict binary $y \in \{-1, +1\}$

$$\text{0-1 Loss}(\cdot) = \begin{cases} 0, & \text{if } \text{sign}(f_{\theta}(x)) = \text{sign}(y) \\ 1, & \text{if } \text{sign}(f_{\theta}(x)) \neq \text{sign}(y) \end{cases}$$

e.g., $y_1 = -1, f_{\theta}(x_1) = -1000.5 \Rightarrow \text{Correct! 0-loss}$

$$y_2 = +1, f_{\theta}(x_2) = -1$$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Classification : predict binary $y \in \{-1, +1\}$

$$\text{0-1 Loss}(\cdot) = \begin{cases} 0, & \text{if } \text{sign}(f_{\theta}(x)) = \text{sign}(y) \\ 1, & \text{if } \text{sign}(f_{\theta}(x)) \neq \text{sign}(y) \end{cases}$$

e.g., $y_1 = -1, f_{\theta}(x_1) = -1000.5 \Rightarrow \text{Correct! 0-loss}$

$y_2 = +1, f_{\theta}(x_2) = -1 \Rightarrow \text{Wrong! 1-loss}$

Optimization under Uncertainty: ML

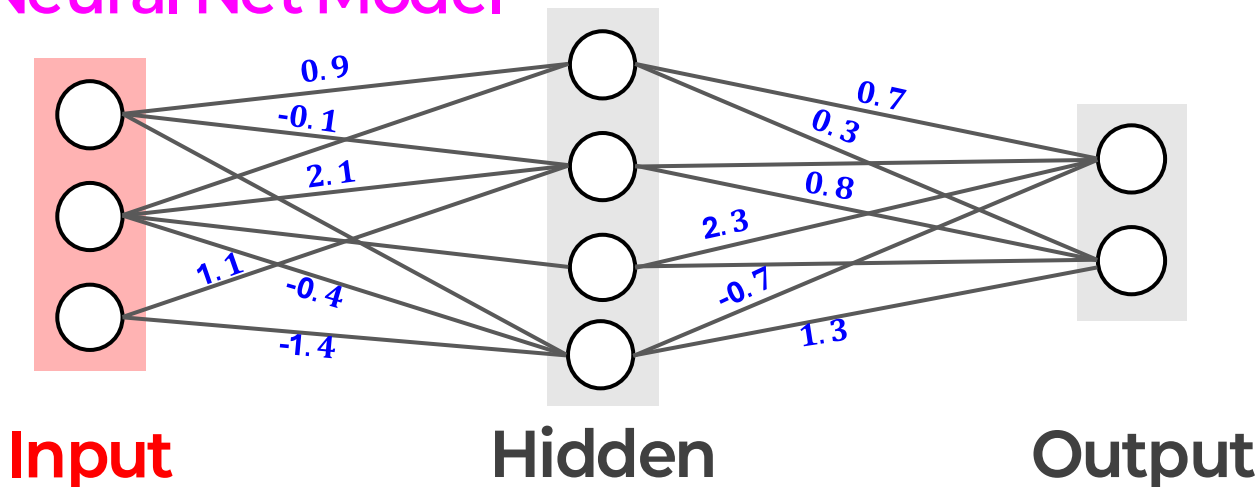
Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Linear Model

$$f_{\theta}(\tilde{x}) = \theta^{\top} \mathbf{x} = \sum_i x_i \theta_i$$

Neural Net Model



Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

ML $\subset$ optimization under uncertainty

➤ OR :

$$\begin{aligned} & \min_x \mathbb{E}_{\mathbb{P}} [f(x, \tilde{\xi})] \\ & \text{s.t. } \text{Prob}[g(x, \tilde{\xi}) \leq 0] \geq \alpha \end{aligned}$$

➤ Control

$$\begin{aligned} & \min_{u_1, u_2, \dots, u_{T-1}} \mathbb{E}_{\mathbb{P}} [\sum_{t=1}^{T-1} c_t(\tilde{x}_t, u_t, \tilde{\xi}_t) + c_T(\tilde{x}_T)] \\ & \text{s.t. } \tilde{x}_{t+1} = f_t(\tilde{x}_t, u_t, \tilde{\xi}_t) \quad \forall t = 1, \dots, T-1 \end{aligned}$$

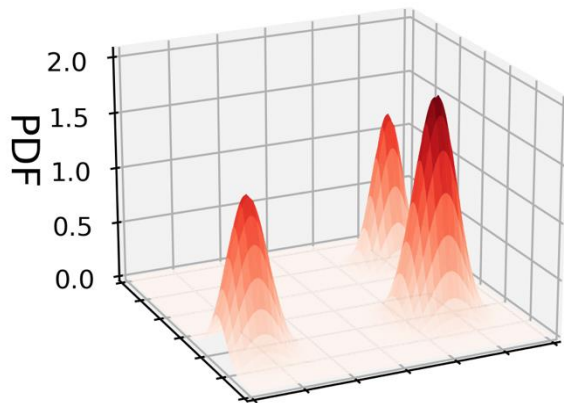
Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

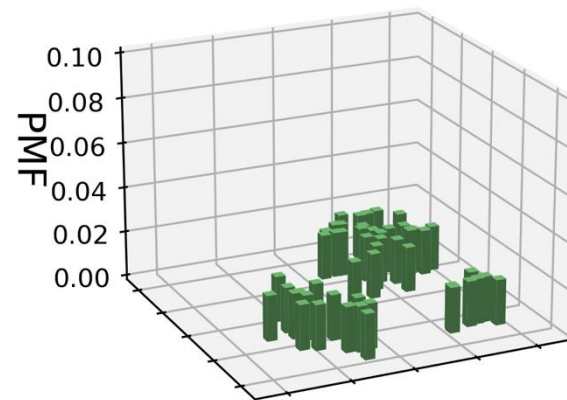
$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

$\mathbb{P}$ unknown $\Rightarrow \theta^*$ not available

Only N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ available: $(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix})$



True distribution $\mathbb{P}$



Empirical distribution $\hat{\mathbb{P}}_N$

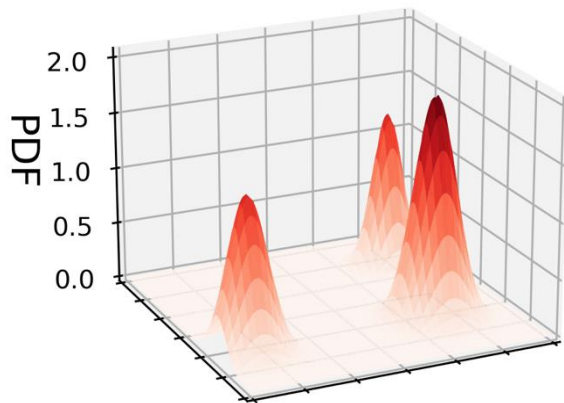
Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

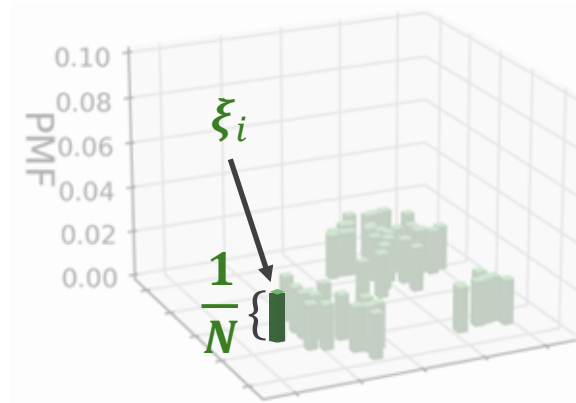
$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

$\mathbb{P}$ unknown $\Rightarrow \theta^*$ not available

Only N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ available: $(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix})$



True distribution $\mathbb{P}$



Empirical distribution $\hat{\mathbb{P}}_N$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter θ^* is

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))]$$

Data-Driven Optimization

Assume $\xi \sim$ empirical distribution $\hat{\mathbb{P}}_N$, then the best empirical parameter $\hat{\theta}_N$ is

$$\begin{aligned} \hat{\theta}_N &\in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\tilde{y}, f_{\theta}(\tilde{x}))] \\ &= \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(y_i, f_{\theta}(x_i)) \end{aligned}$$

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

$\neq$

$$\hat{\theta}_N \in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

Data-driven Problem

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

$\neq$

$$\hat{\theta}_N \in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

Data-driven Problem

If $\hat{\mathbb{P}}_N \approx \mathbb{P}$, then $\hat{\theta}_N$ is a good approximation to θ^*

- In ML, $\hat{\theta}_N$ is called Empirical Risk Minimizer (ERM)
- In OR, it is called Sample Average Approximation (SAA)

Optimization under Uncertainty: ML

$$\theta^* \neq \hat{\theta}_N !$$

If $\hat{\mathbb{P}}_N$ poorly approximates $\mathbb{P}$ —especially with sparse data—

$\hat{\theta}_N$ may perform poorly.

Optimization under Uncertainty: ML

True Problem

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathbb{P}} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

$\neq$

$$\hat{\theta}_N \in \arg \min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

Data-driven Problem

If $\hat{\mathbb{P}}_N$ poorly approximates $\mathbb{P}$ —especially with sparse data—

$\hat{\theta}_N$ may perform poorly.

Distributionally Robust Optimization (DRO)

$$\min_{\theta} \mathbb{E}_{\hat{\mathbb{P}}_N} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

Distributionally Robust Optimization (DRO)

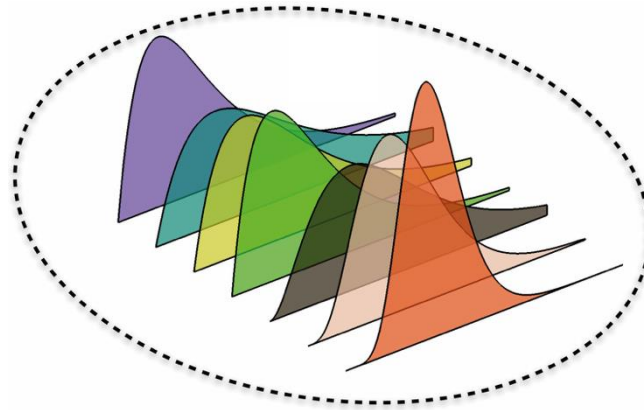
$$\min_{\theta} \max_{Q \in \mathcal{P}_{\lambda}(\hat{\mathbb{P}}_N)} \mathbb{E} [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

Distributionally Robust Optimization (DRO)

$$\min_{\theta} \max_{Q \in \mathcal{P}_{\lambda}(\hat{\mathbb{P}}_N)} \mathbb{E}_Q [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$

Distributionally Robust Optimization (DRO)

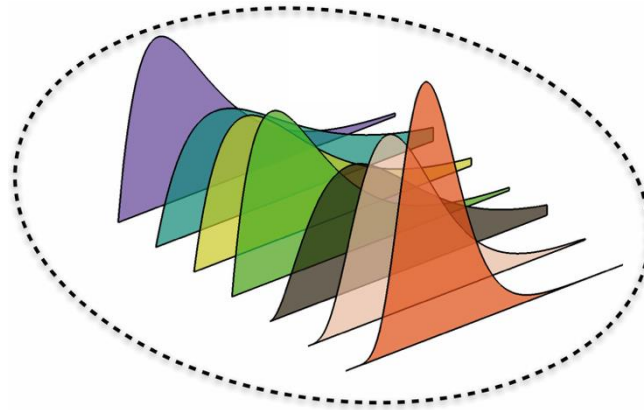
$$\min_{\theta} \max_{Q \in \mathcal{P}_{\lambda}(\hat{\mathbb{P}}_N)} \mathbb{E}_Q [\ell(\tilde{\mathbf{y}}, f_{\theta}(\tilde{\mathbf{x}}))]$$



$\mathcal{P}_{\lambda}(\cdot)$: Ambiguity set

Distributionally Robust Optimization (DRO)

$$\hat{\theta}_N^{DRO} \in \arg \min_{\theta} \max_{Q \in \mathcal{P}_\lambda(\hat{\mathbb{P}}_N)} \mathbb{E}_Q [\ell(\tilde{\mathbf{y}}, f_\theta(\tilde{\mathbf{x}}))]$$

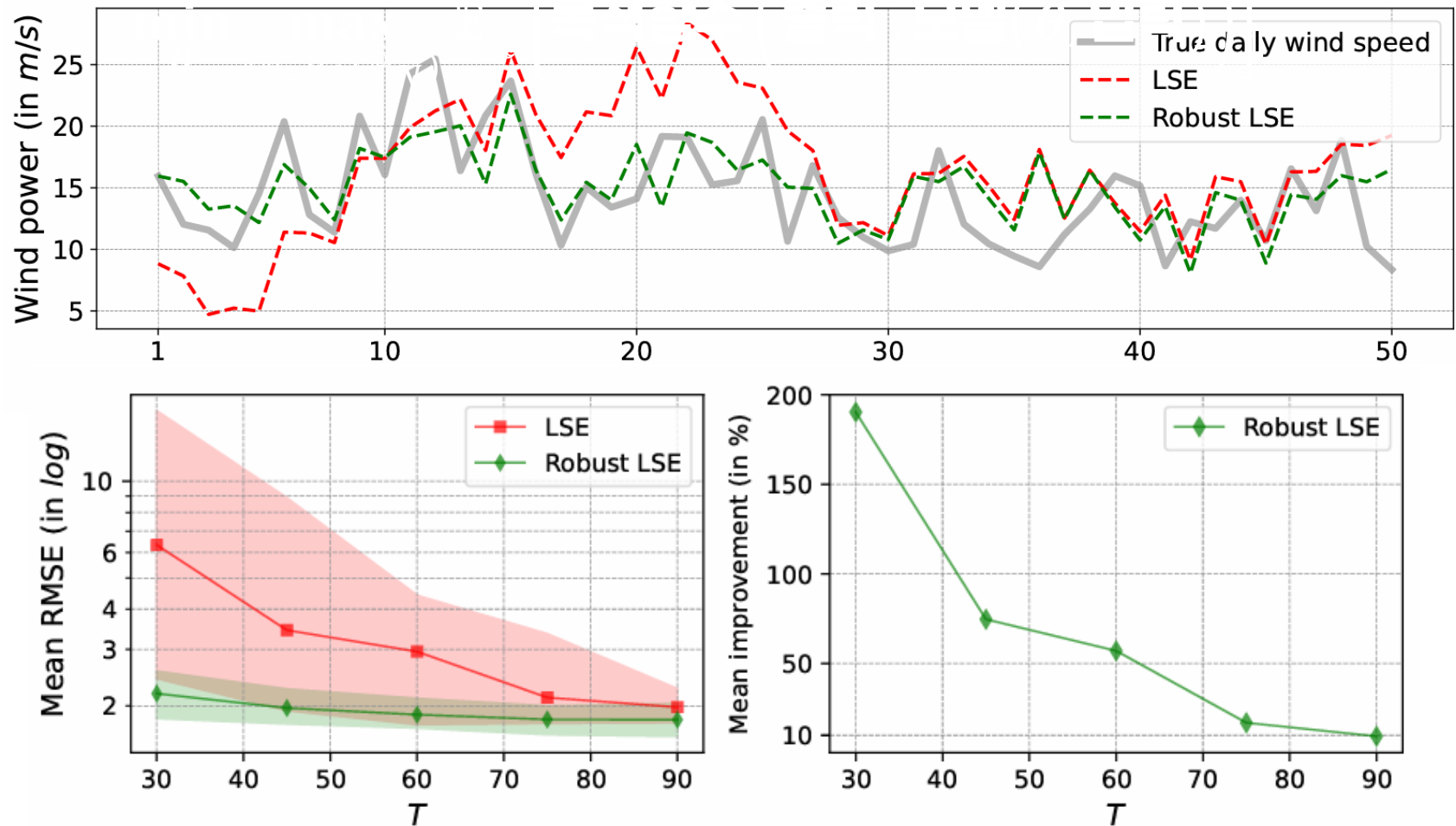


$\mathcal{P}_\lambda(\cdot)$: Ambiguity set

$\hat{\theta}_N^{DRO} \Rightarrow$ Robust performance under data scarcity

Machine Learning

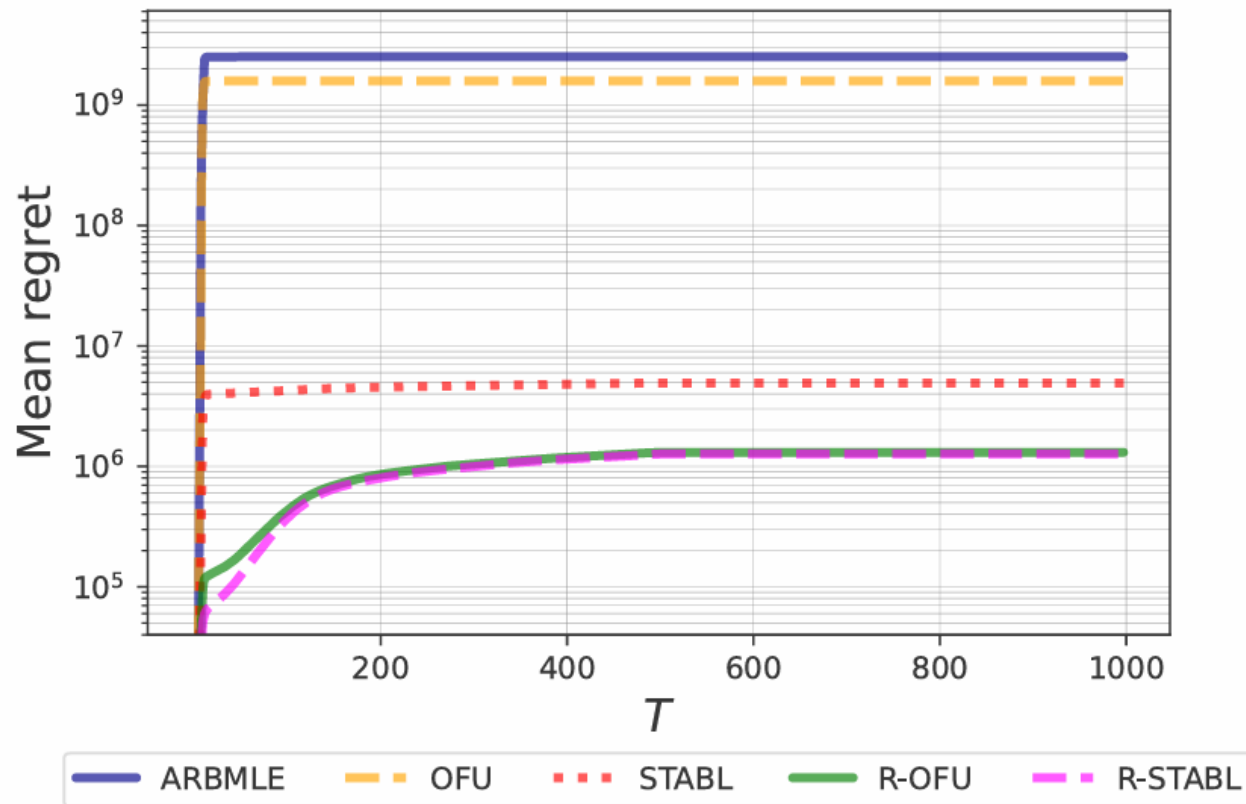
DRO-based NN model : Short-term Wind Prediction ⁴⁾



⁴⁾ Park et al., "Robust System Identification", *ICLR*, 2025

Machine Learning

DRO-based Reinforcement Learning : Boeing 747 Control ⁴⁾



⁴⁾ Park et al., "Robust System Identification", *ICLR*, 2025

Operations Research

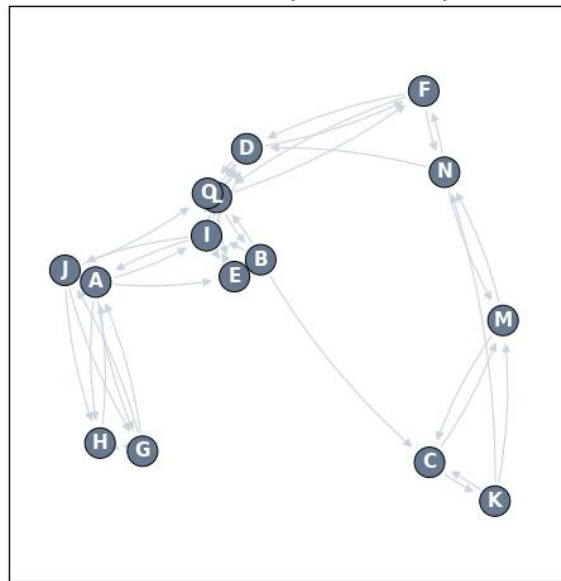
DRO SDDP : Portfolio Optimization ⁵⁾

Dataset	Scheme	Utility	Mean return	Std. Dev.	Sharpe ratio
10 Industry Fama-French	DDR-SDDP	3.19E-02	3.41E-02	6.35E-02	4.77E-01
	DD-SDDP	2.65E-02	2.90E-02	6.95E-02	3.62E-01
	Independent	1.58E-02	1.76E-02	6.23E-02	2.21E-01
	HMM-SDDP	2.43E-02	2.54E-02	4.70E-02	4.59E-01
	Equal	-2.15E-03	-1.97E-03	2.74E-02	-3.52E-02
12 Industry Fama-French	DDR-SDDP	2.49E-02	2.67E-02	5.97E-02	3.82E-01
	DD-SDDP	2.35E-02	2.59E-02	6.90E-02	3.20E-01
	Independent	2.01E-02	2.20E-02	6.32E-02	2.88E-01
	HMM-SDDP	1.97E-02	2.09E-02	4.97E-02	3.42E-01
	Equal	-1.52E-03	-1.36E-03	2.72E-02	-3.17E-02
12 Industry Fama-French	DDR-SDDP	1.38E-02	1.43E-02	3.61E-02	2.89E-01
	DD-SDDP	7.34E-03	9.45E-03	6.86E-02	8.16E-02
	Independent	3.43E-03	5.34E-03	6.51E-02	2.28E-02
	HMM-SDDP	5.68E-03	6.09E-03	3.43E-02	6.53E-02
	Equal	-5.00E-02	-4.87E-02	2.73E-02	-3.18E-01

Operations Research

DRO Two-stage LAP (Location-Allocation Problem) : Drone Scheduling

Candidate network & transport costs — press Solve



Two-stage LAP (Sample Average Approximation)

Stage 1 (here-and-now, before the disaster)

$$\min_{\mathbf{x}, \mathbf{r}} \sum_j c_j x_j + \sum_j f_j r_j + \frac{1}{N} \sum_{n=1}^N Q(\mathbf{x}, \mathbf{r}; \hat{\xi}^n)$$

$$\text{s.t. } \sum_{j \in \mathcal{N}'} r_j \leq B, \quad \sum_{j \in \mathcal{N}'} x_j \leq p$$

$$r_j \leq B x_j, x_j \in \{0, 1\}, r_j \geq 0 \quad \forall j \in \mathcal{N}'$$

Stage 2 (recourse for scenario n, after demand is seen)

$$Q(\mathbf{x}, \mathbf{r}; \hat{\xi}^n) = \min_{\mathbf{y}, \mathbf{q}, \mathbf{s}} \sum_a d_a y_a + \sum_j u_j q_j + \sum_j o_j s_j$$

$$\text{s.t. } (\text{in}_j - \text{out}_j) + q_j - s_j = \hat{\xi}_j^n - r_j \quad \forall j \in \mathcal{N}'$$

$$0 \leq y_a \leq \bar{y} \quad \forall a \in \mathcal{A}; \quad 0 \leq q_j \leq \hat{\xi}_j^n, s_j \geq 0$$

DECISION VARIABLES (what we choose)

s1: $\mathbf{x}$ open facilities, $\mathbf{r}$ reserves; s2: $\mathbf{y}$ flows, $\mathbf{q}$ shortages, $\mathbf{s}$ surpluses

UNCERTAINTY (seen only after stage 1)

$\hat{\xi}^n$ = demand vector in scenario n ($\hat{\xi}_j^n$ at node j)

PARAMETERS (given numbers)

c_j open, f_j reserve, B budget, p cap; d_a transport, u_j short, o_j surplus, N #scen

SETS & INDICES

graph $G = (\mathcal{N}', \mathcal{A})$: $\mathcal{N}'$ nodes, $\mathcal{A}$ arcs; $j \in \mathcal{N}'$, $a \in \mathcal{A}$, $n = 1, \dots, N$



press Solve to choose where to build and how much to stock

trials slider from 1 to 300, currently at 300

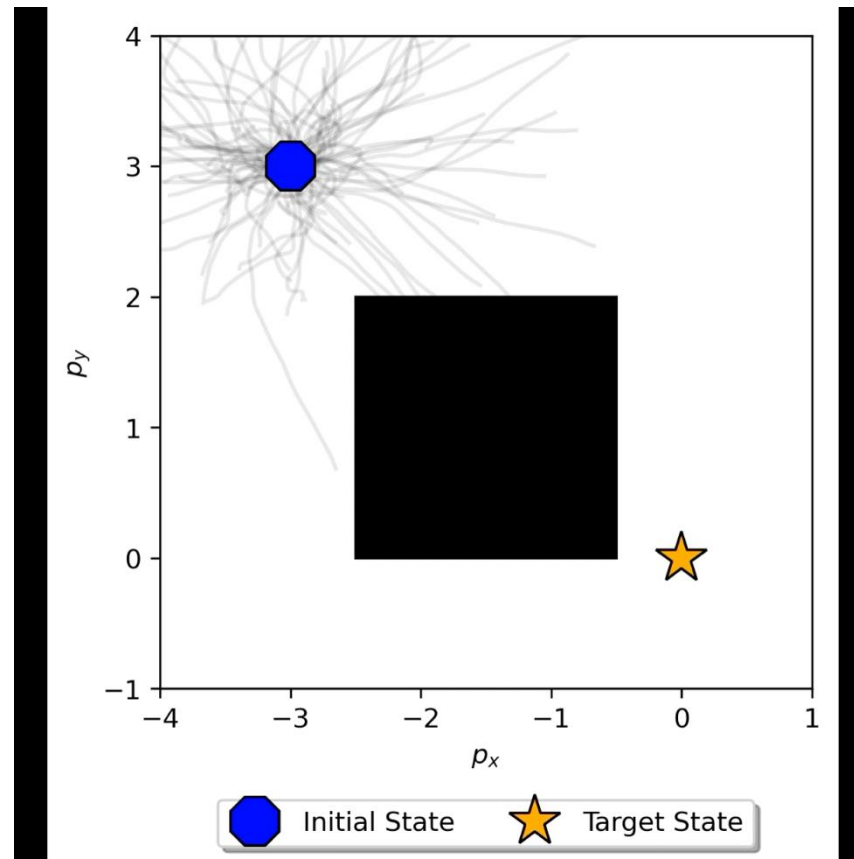
band: IQR 25-75

Run trials

Cost vs N curve

Control

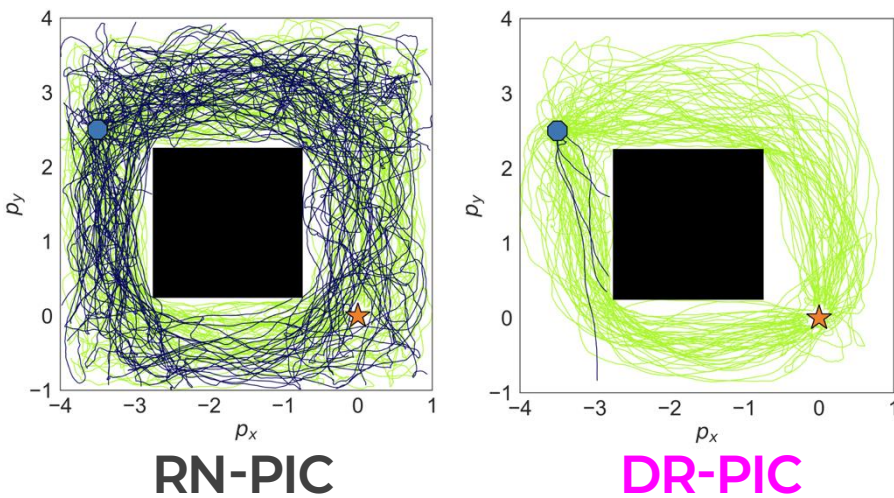
DRO PIC (Path Integral Control) : Robotics ⁶⁾



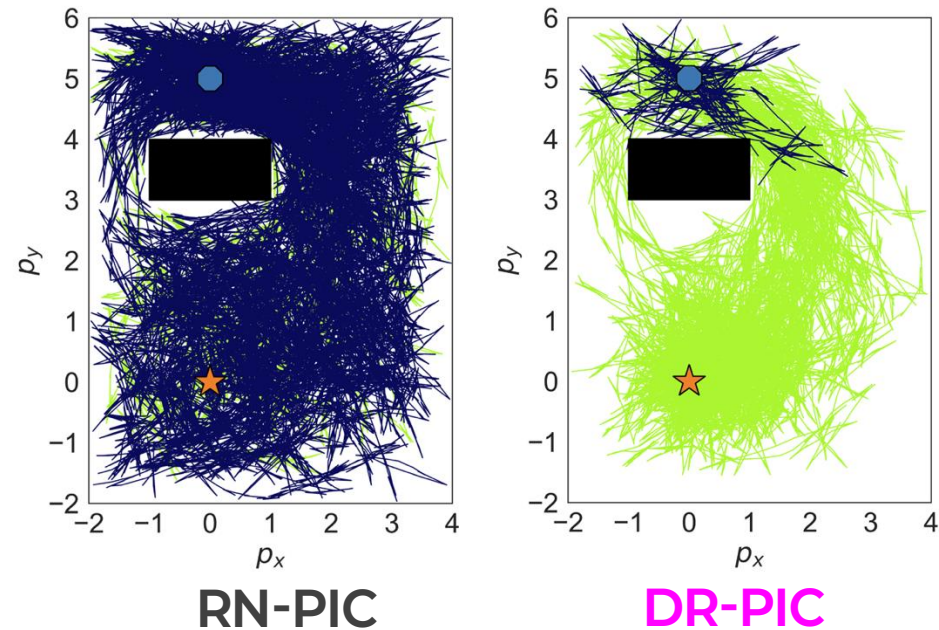
Control

DRO PIC (Path Integral Control) : Robotics ⁶⁾

➤ Particle robot



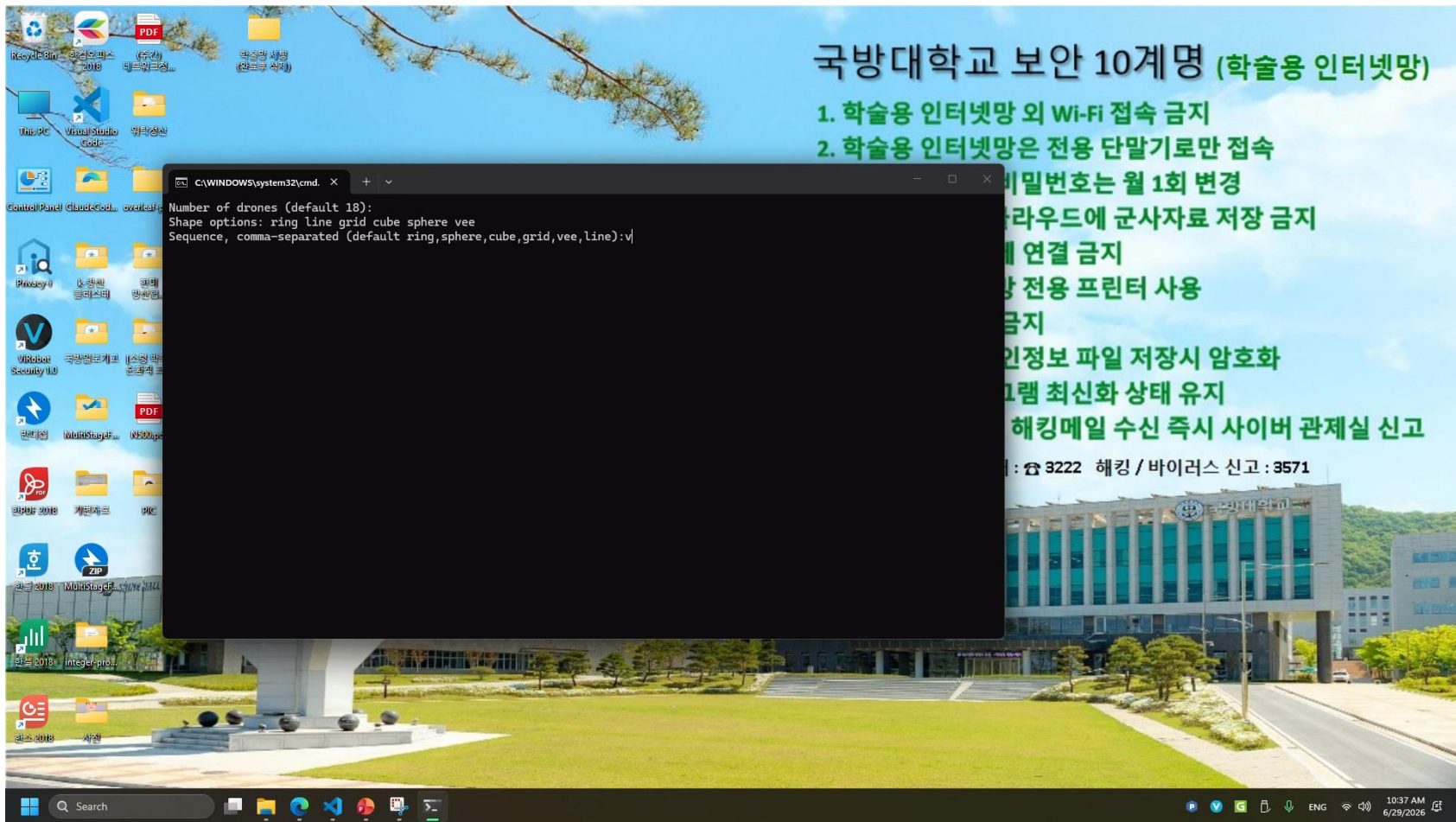
➤ Unicycle



6) Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

Control

DRO PIC (Path Integral Control) : Robotics ⁶⁾



6) Park et al., "Distributionally Robust Path Integral Control," *Proc. ACC*, 2025.

Thanks!
Any question?

